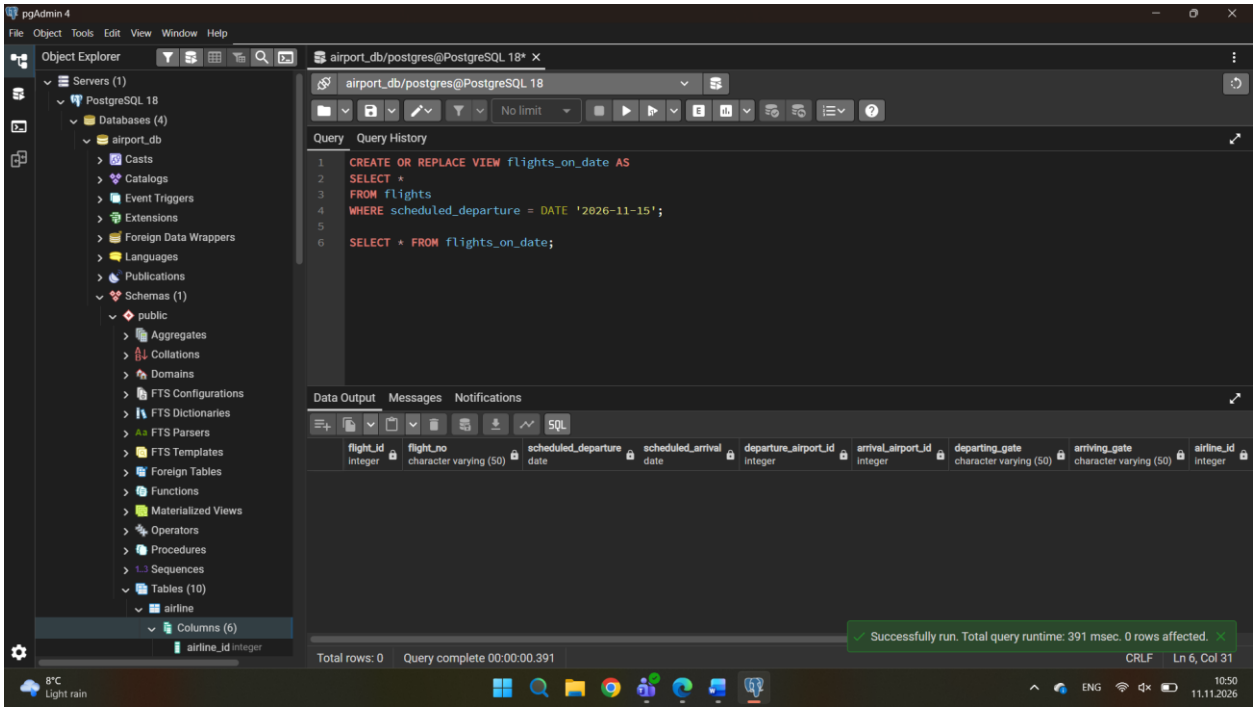
Laboratory work 8

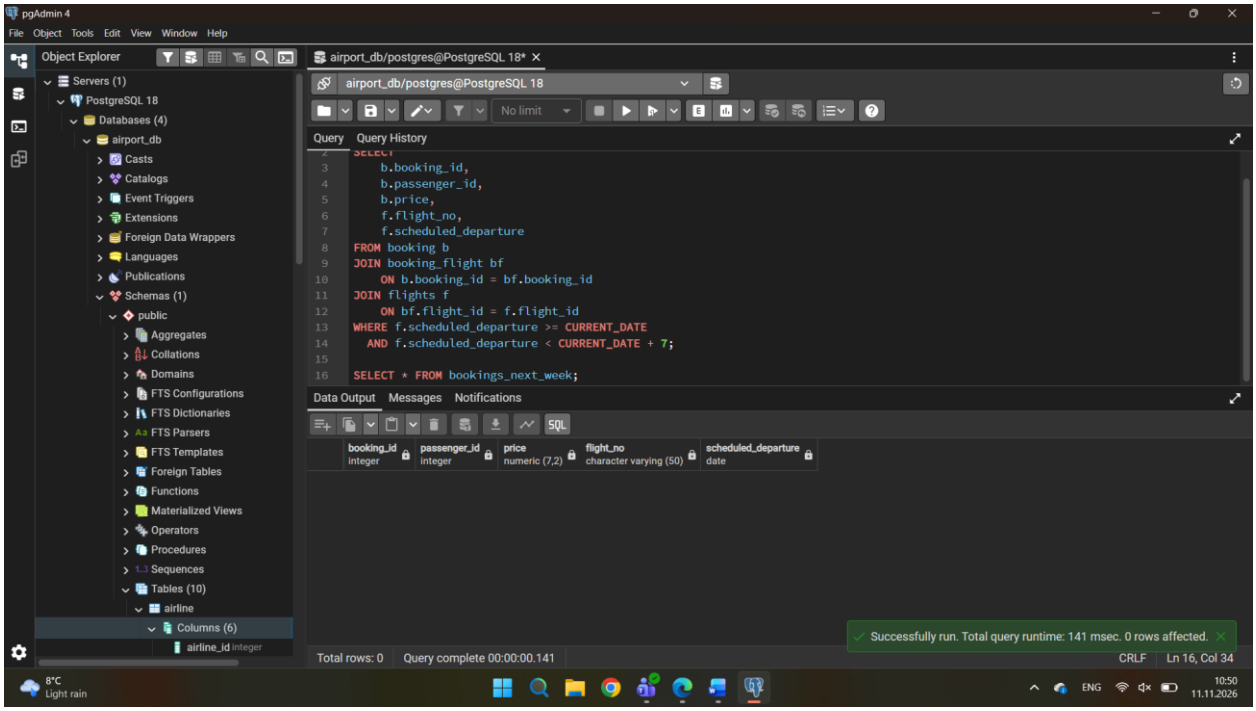
We continue to work with the database from the previous laboratory works
Take a full-page screenshot that covers the code and results of each task.

VIEW

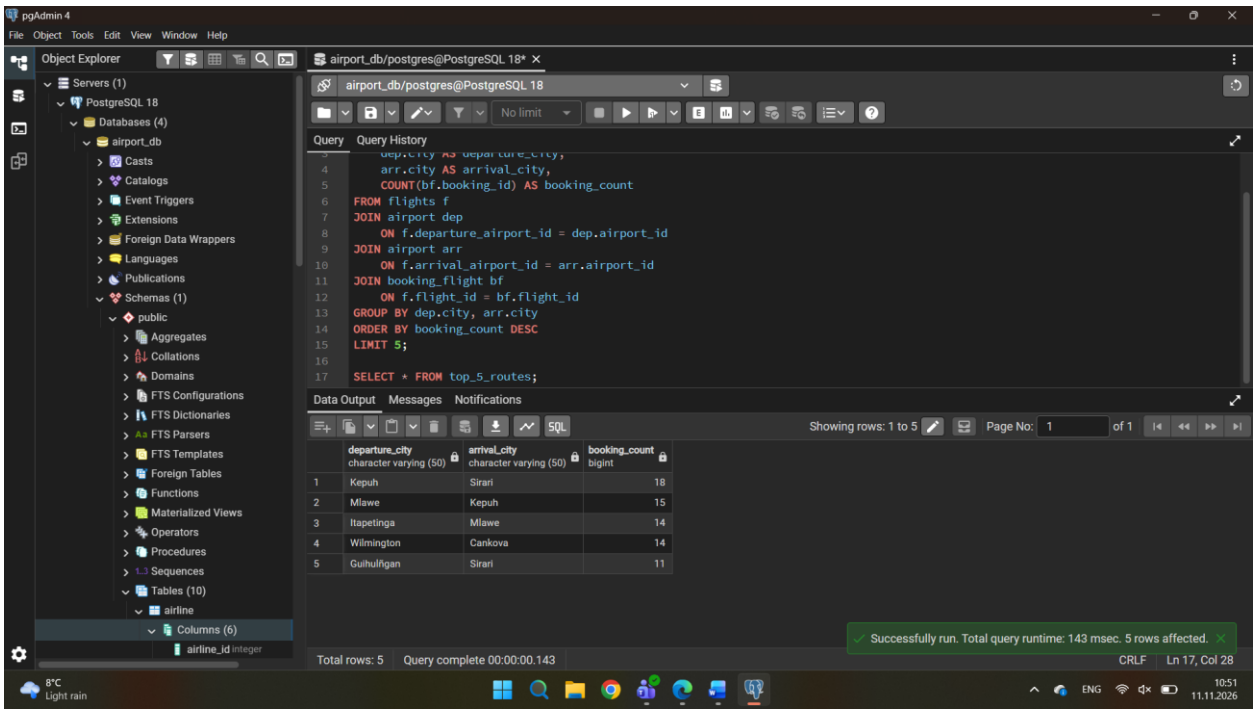
1. Create a view to show details of all flights that are departing on a specific date.



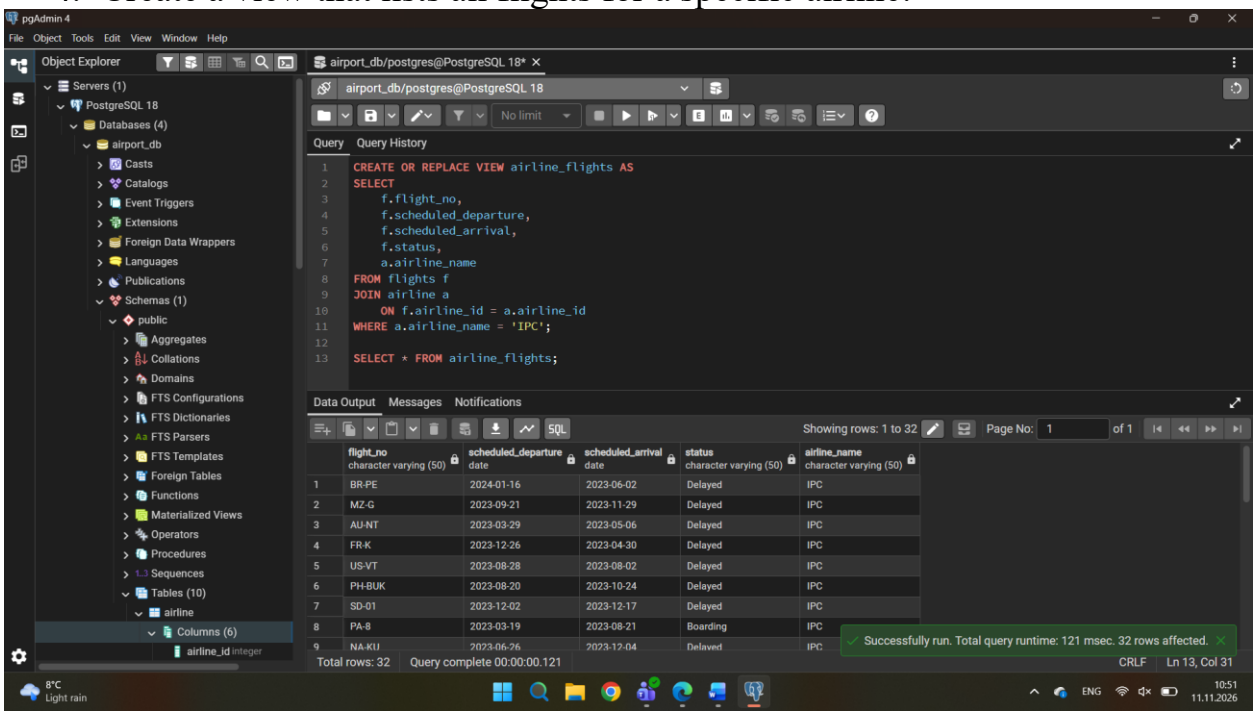
2. Create a view that shows bookings for flights scheduled to depart within the next week.



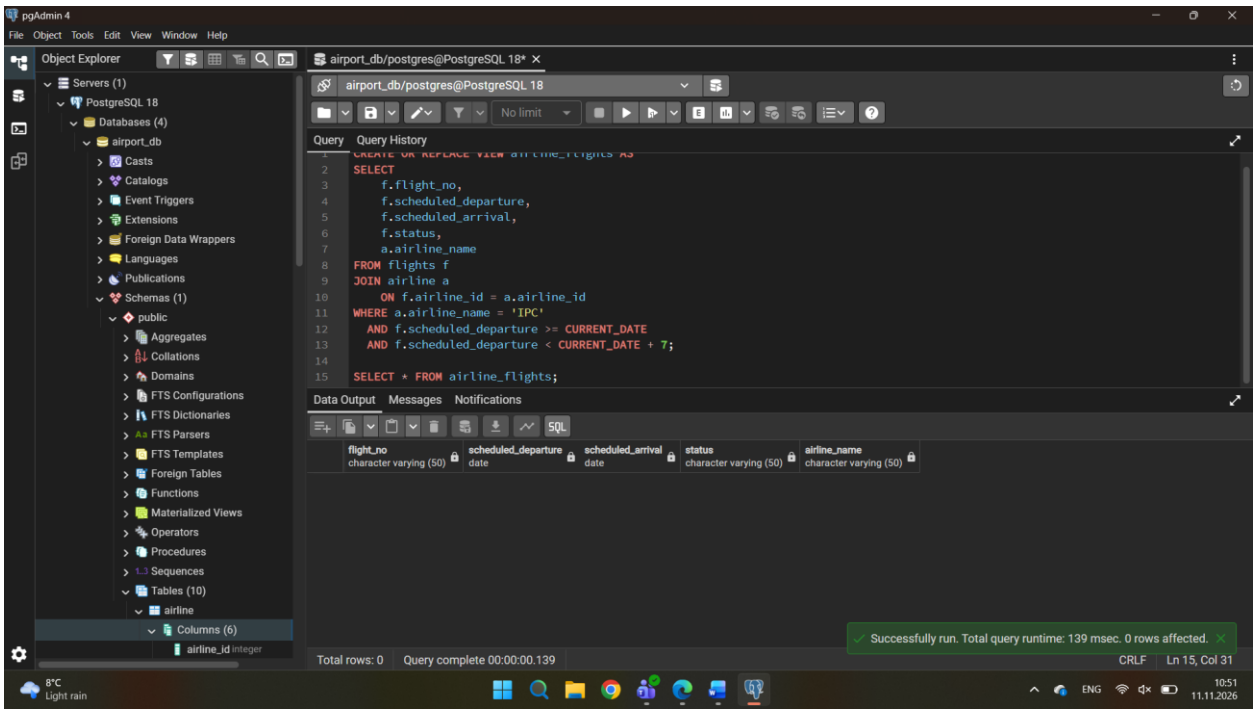
3. Create a view to show the top 5 most popular flight routes based on the number of bookings.



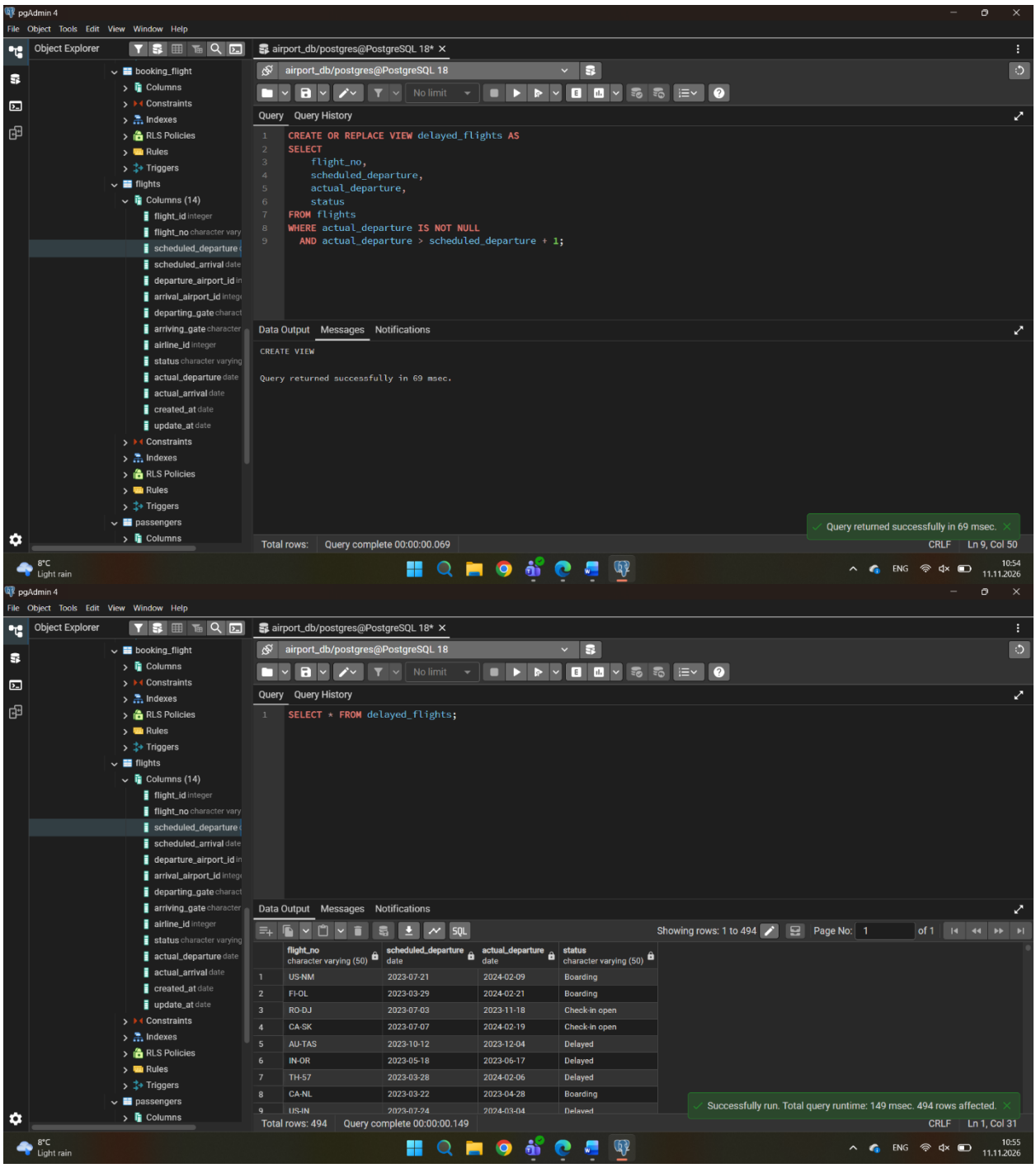
4. Create a view that lists all flights for a specific airline.



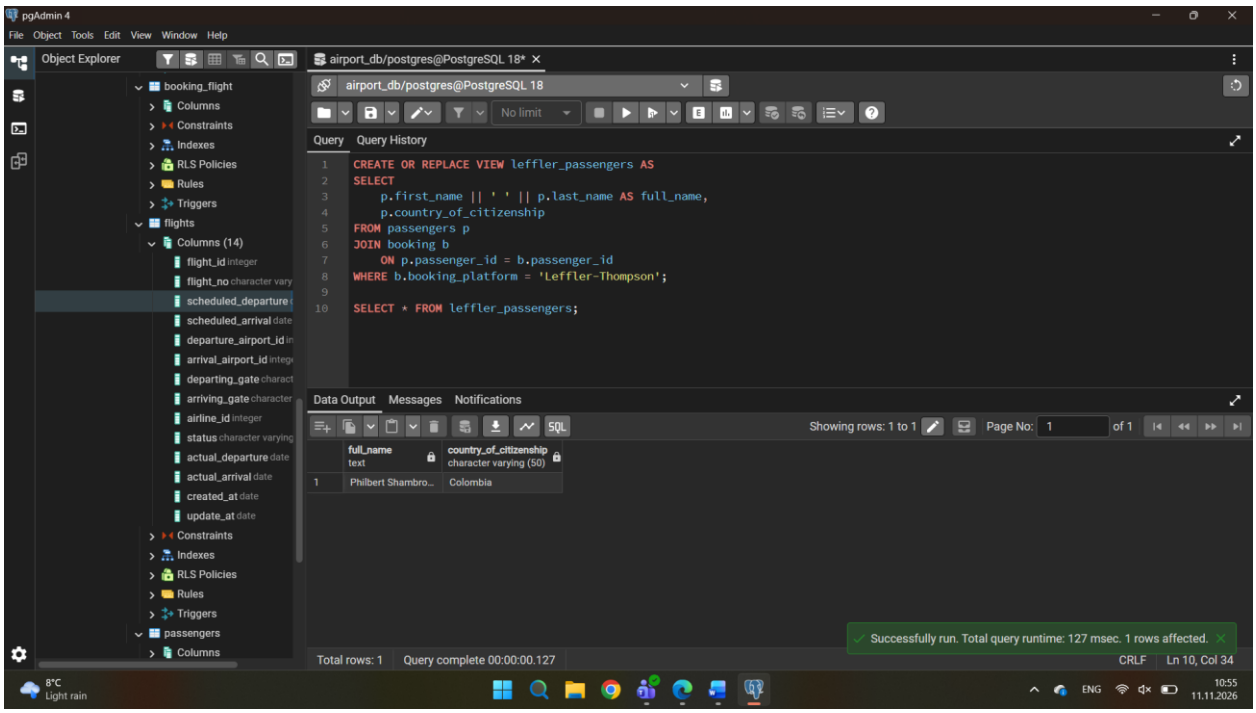
5. Modify the view created in task 4 to show only flights departing within the next 7 days for a specific airline.



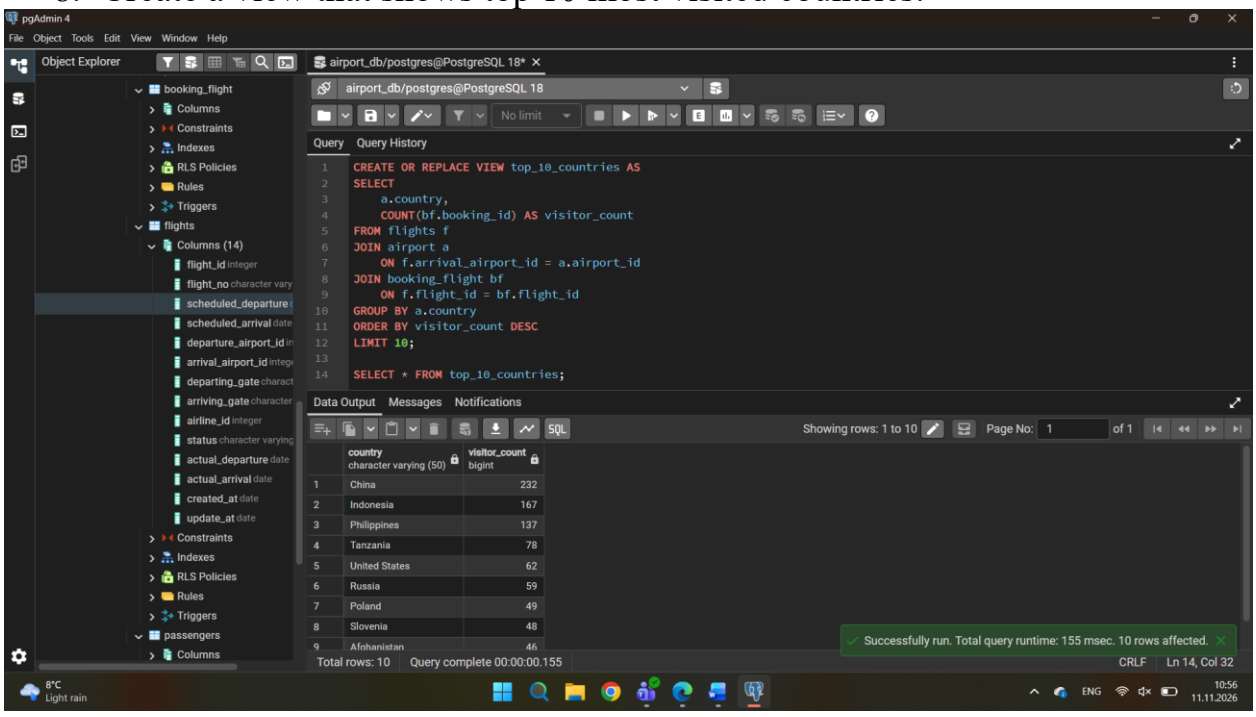
6. Create a view to show flights that are delayed by more than 24 hours.



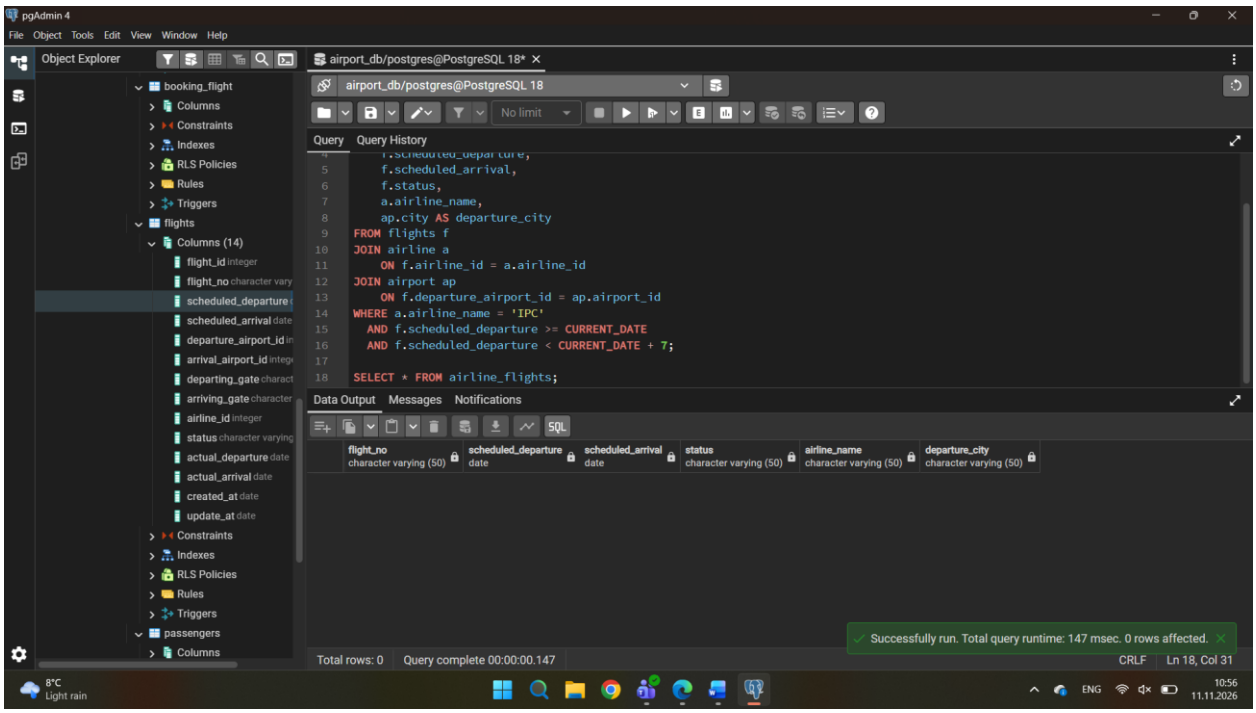
7. Create a view in which you can display the full name and country of origin of passengers who made bookings on Leffler-Thompson platform. Then show the list of those passengers.



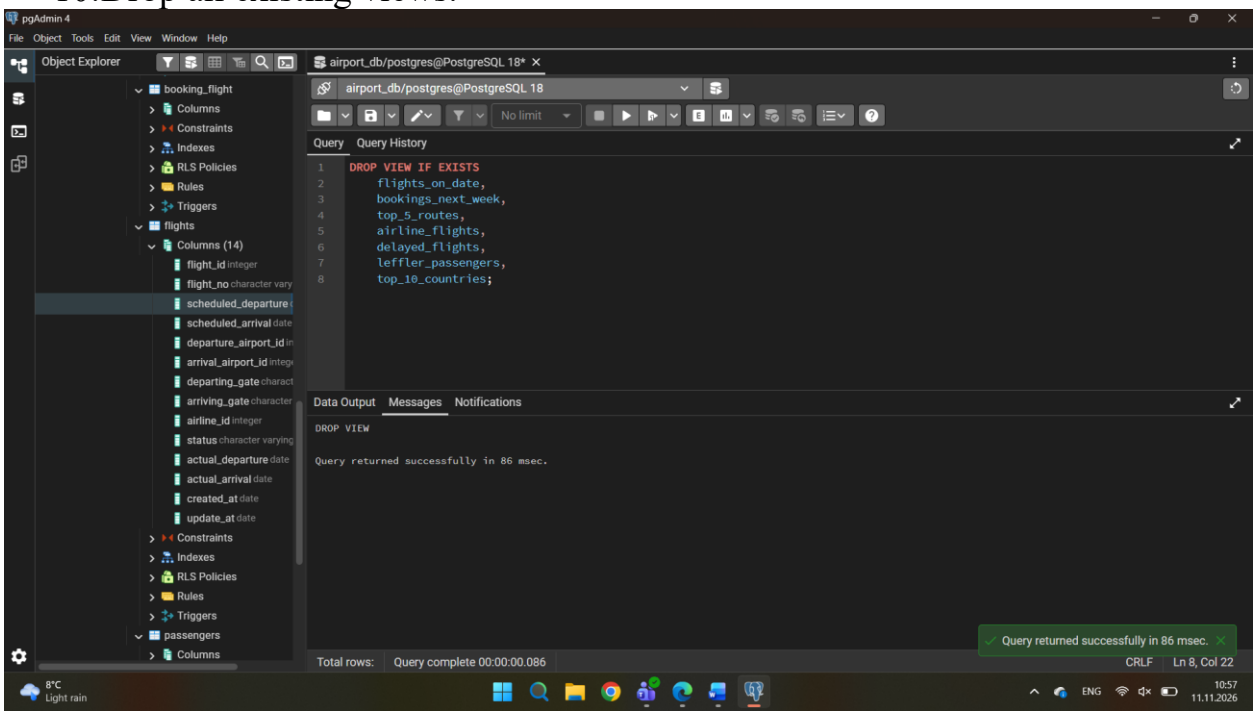
8. Create a view that shows top 10 most visited countries.



9. Update any of the created views by adding new information in the view table. Show results.

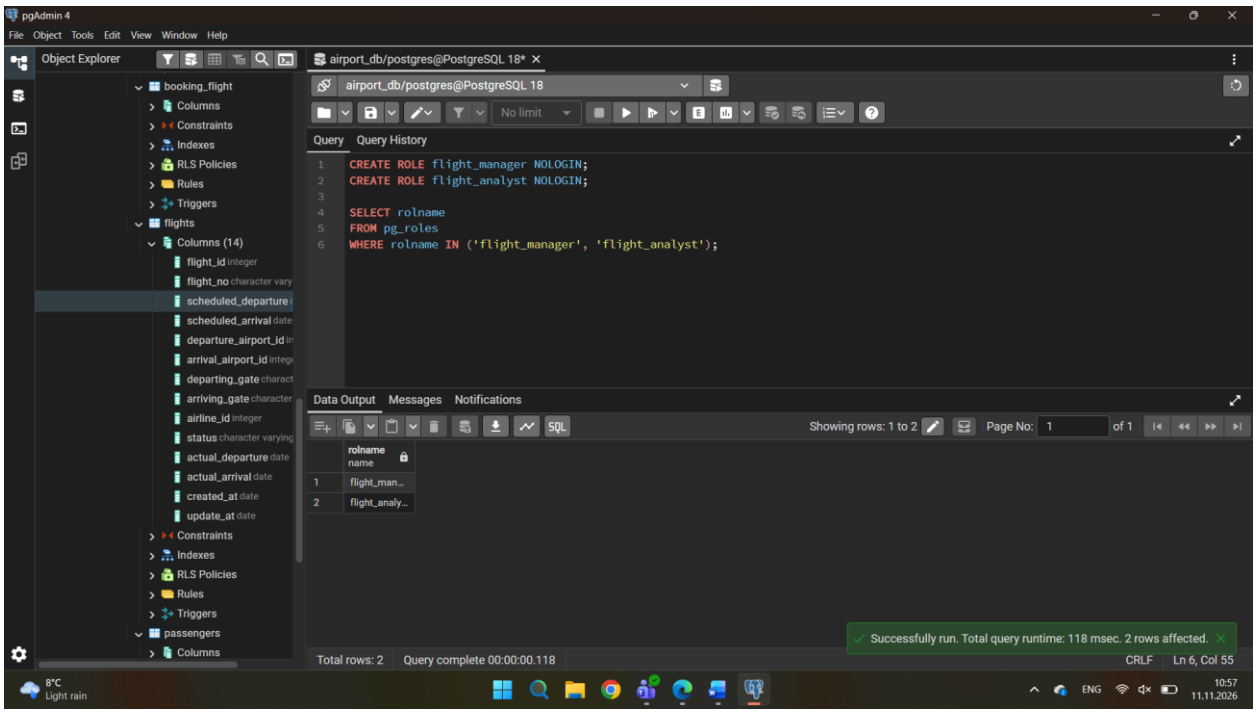


10.Drop all existing views.

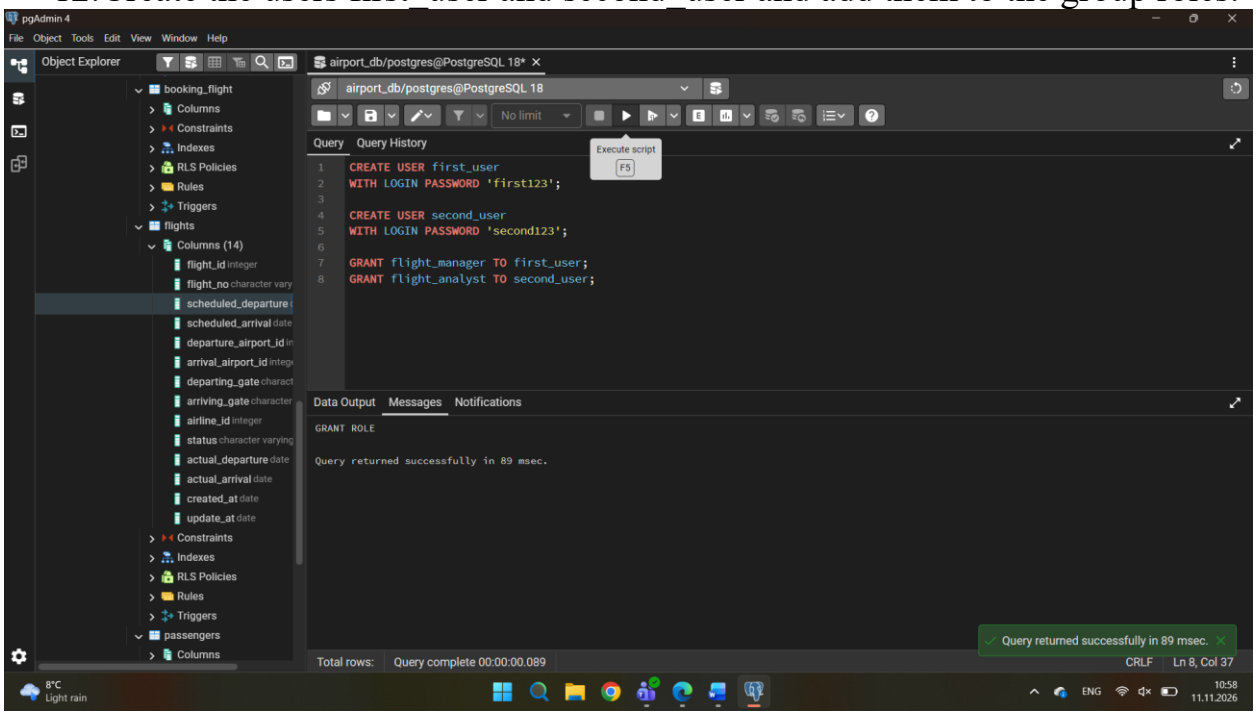


ROLES.

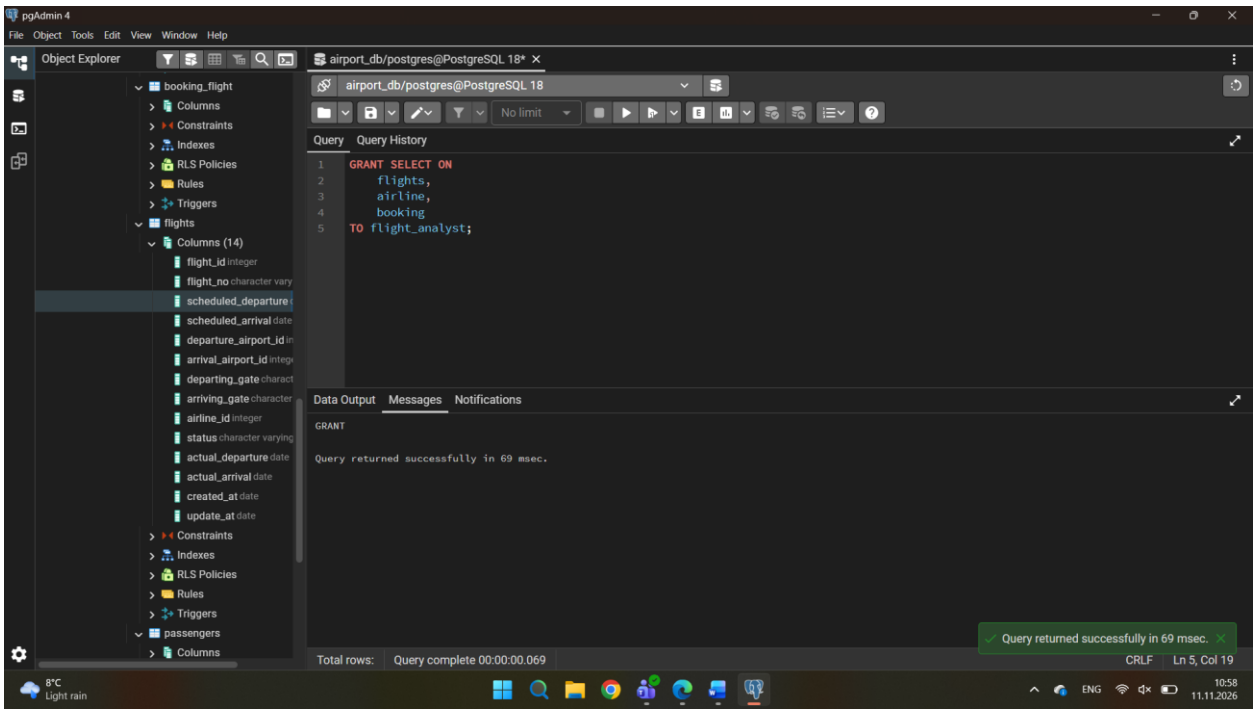
11.Create the group roles flight_manager and flight_analyst, then show the list of roles.



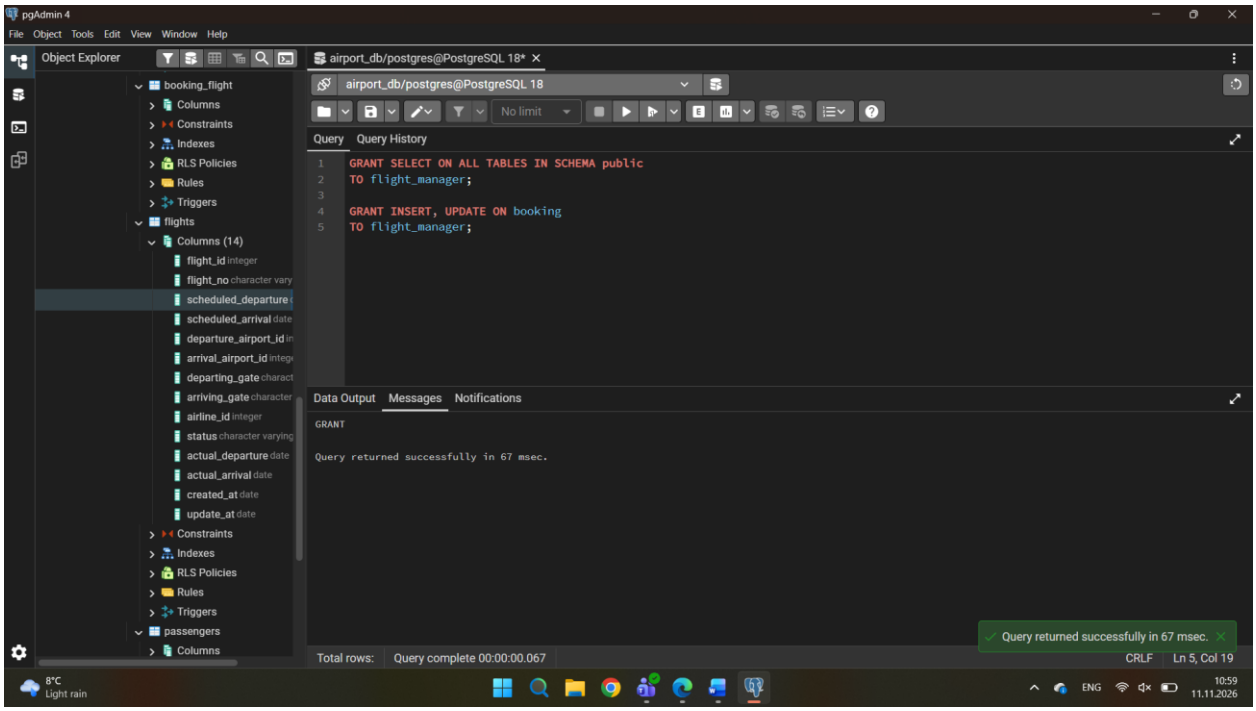
12.Create the users first_user and second_user and add them to the group roles.



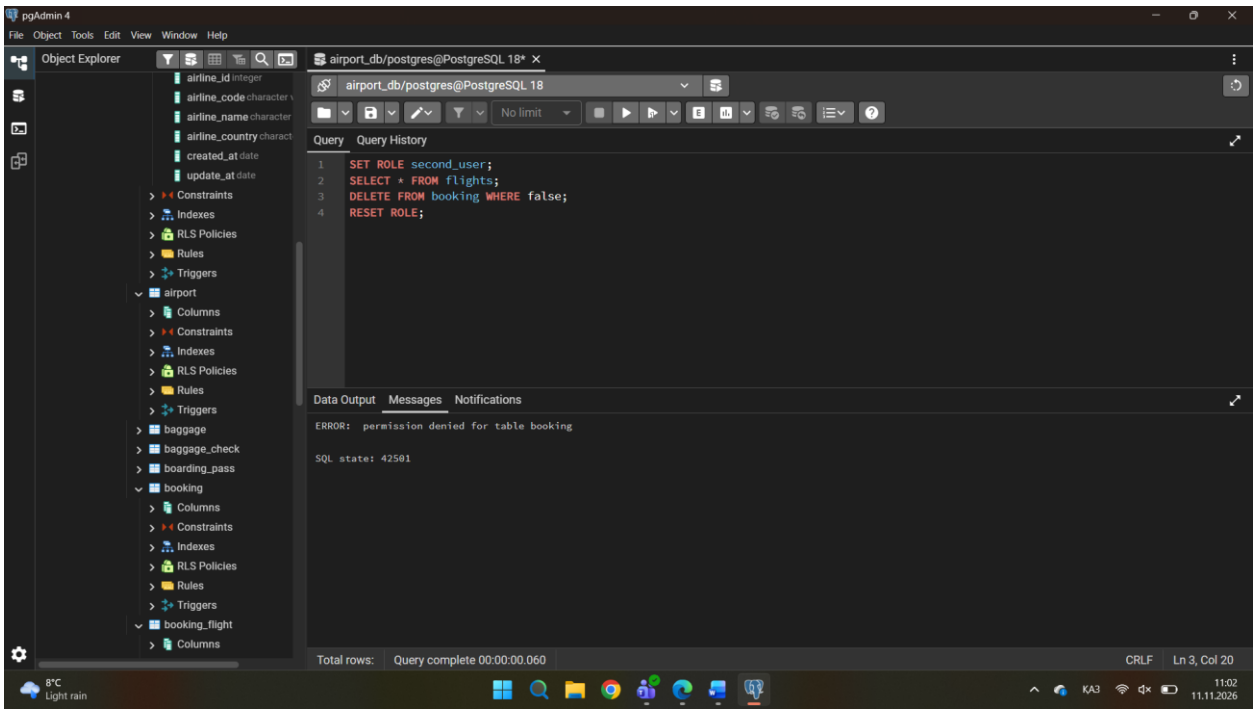
13.Grant flight_analyst SELECT on the flights, airlines and bookings tables.



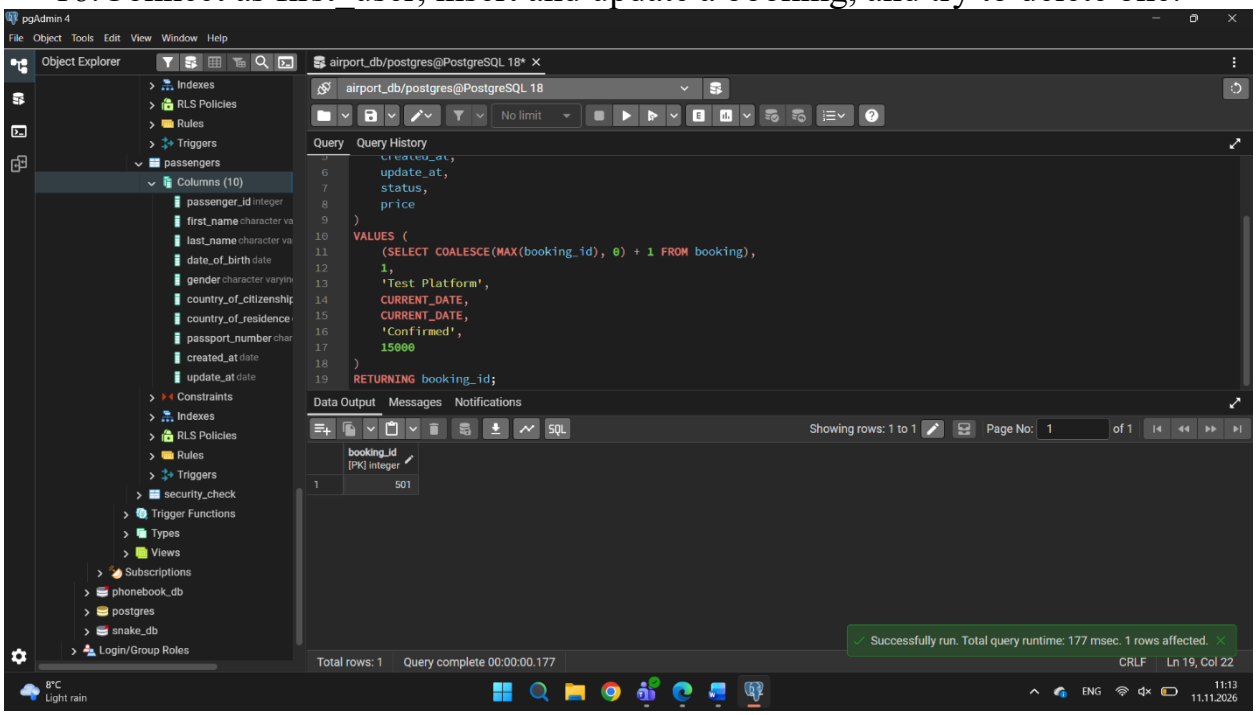
14. Grant flight_manager SELECT, INSERT and UPDATE on bookings, and SELECTon the other tables.

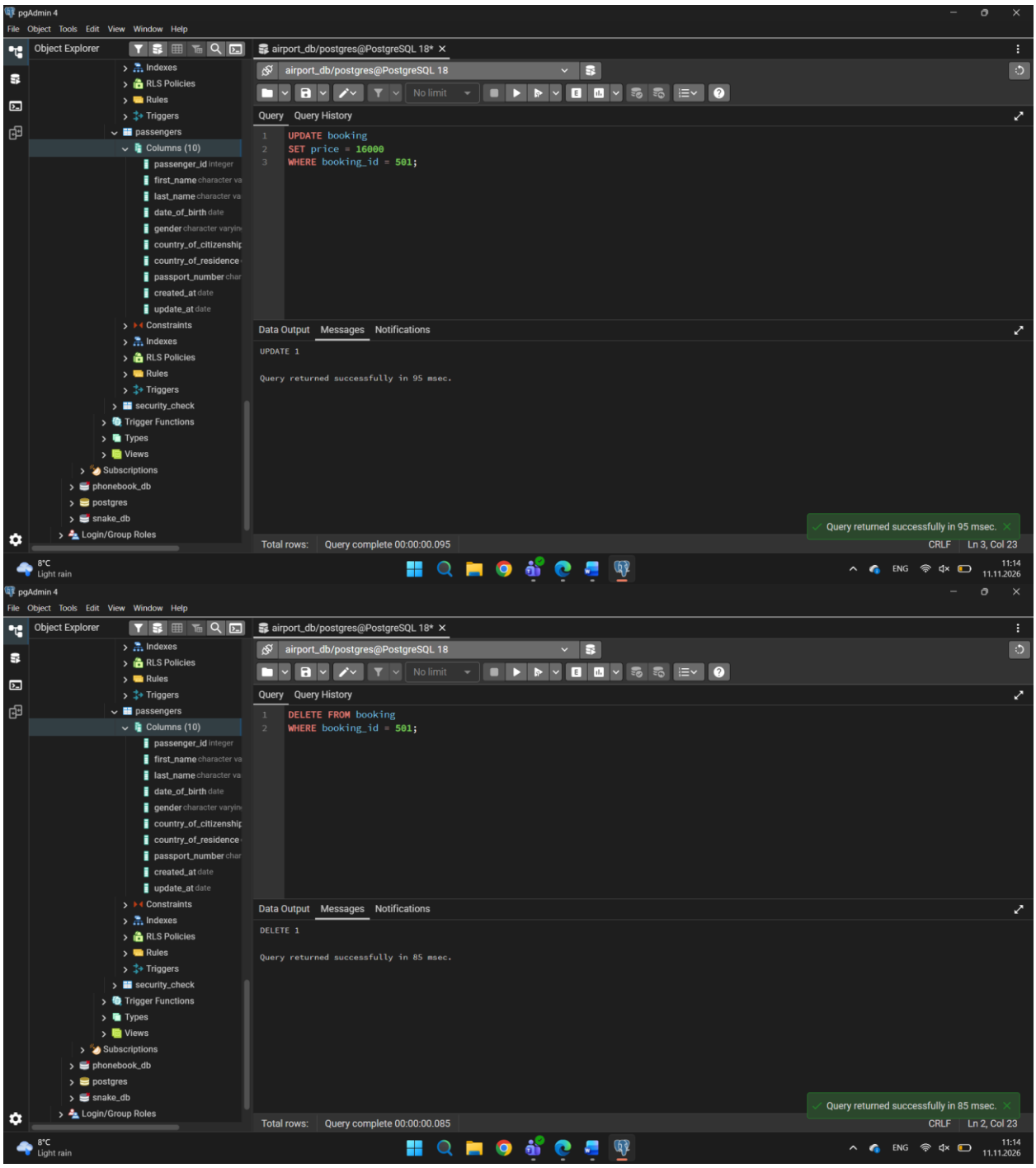


15. Connect as second_user, read flights and try to delete from bookings

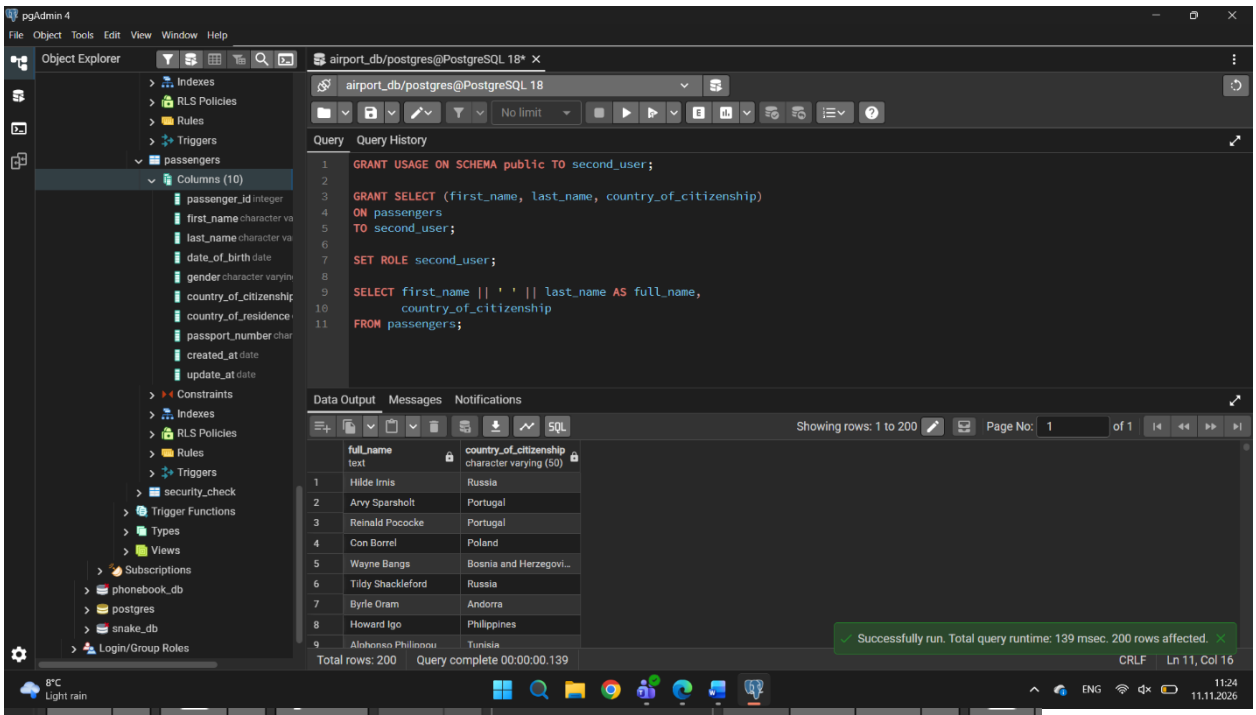


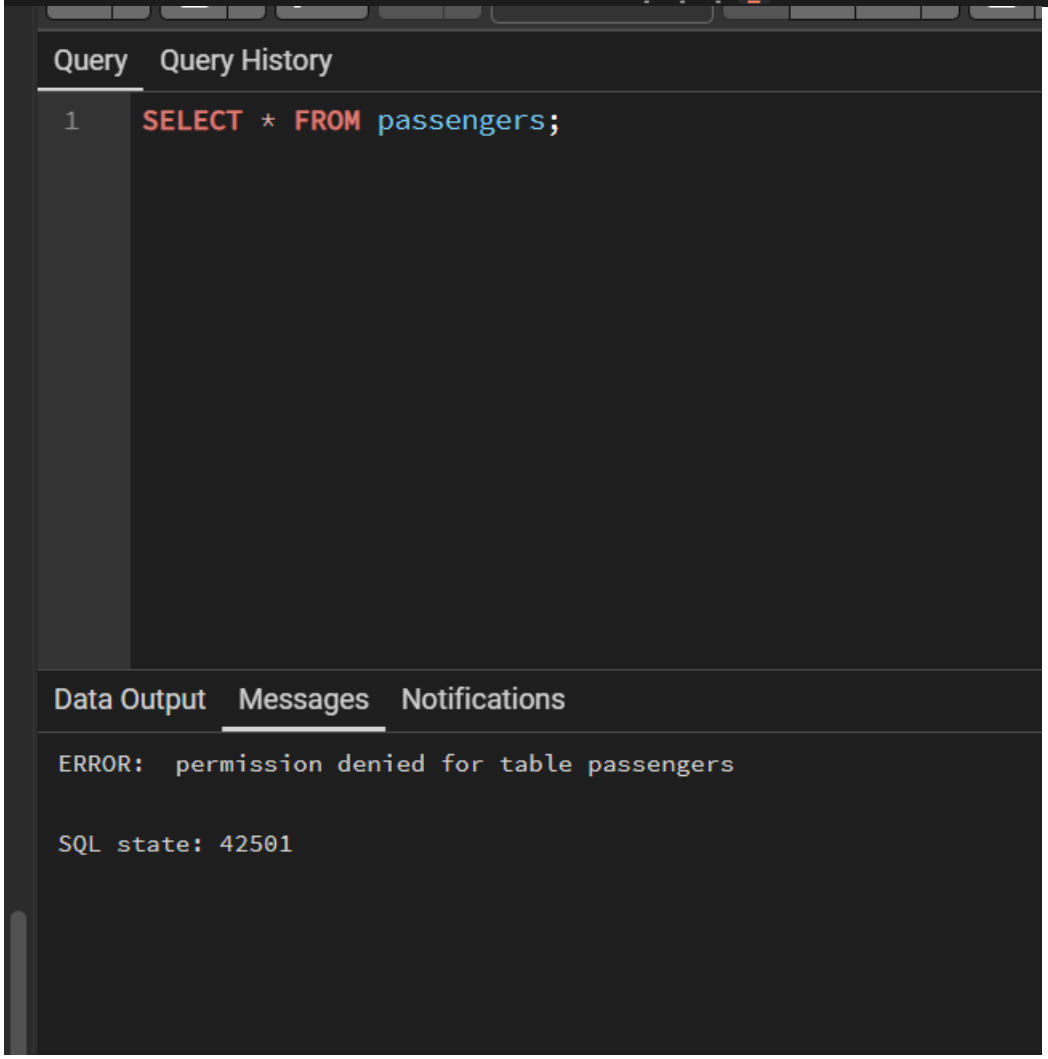
16.Connect as first_user, insert and update a booking, and try to delete one.



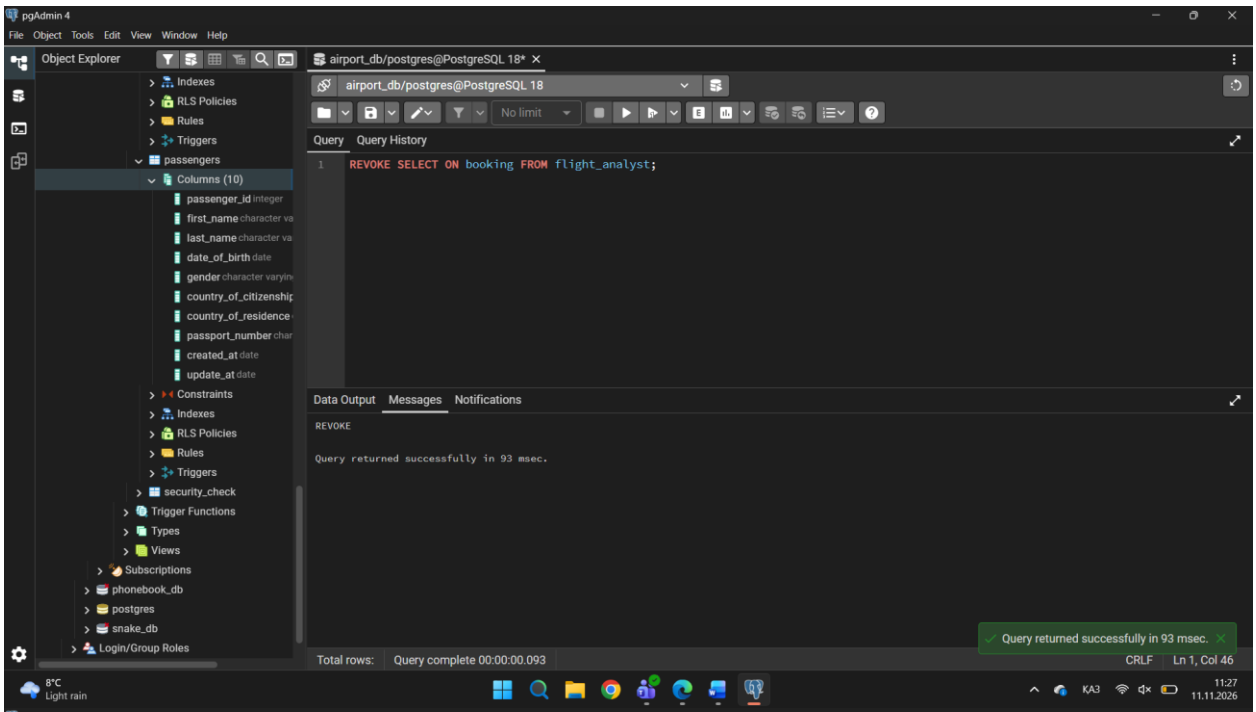


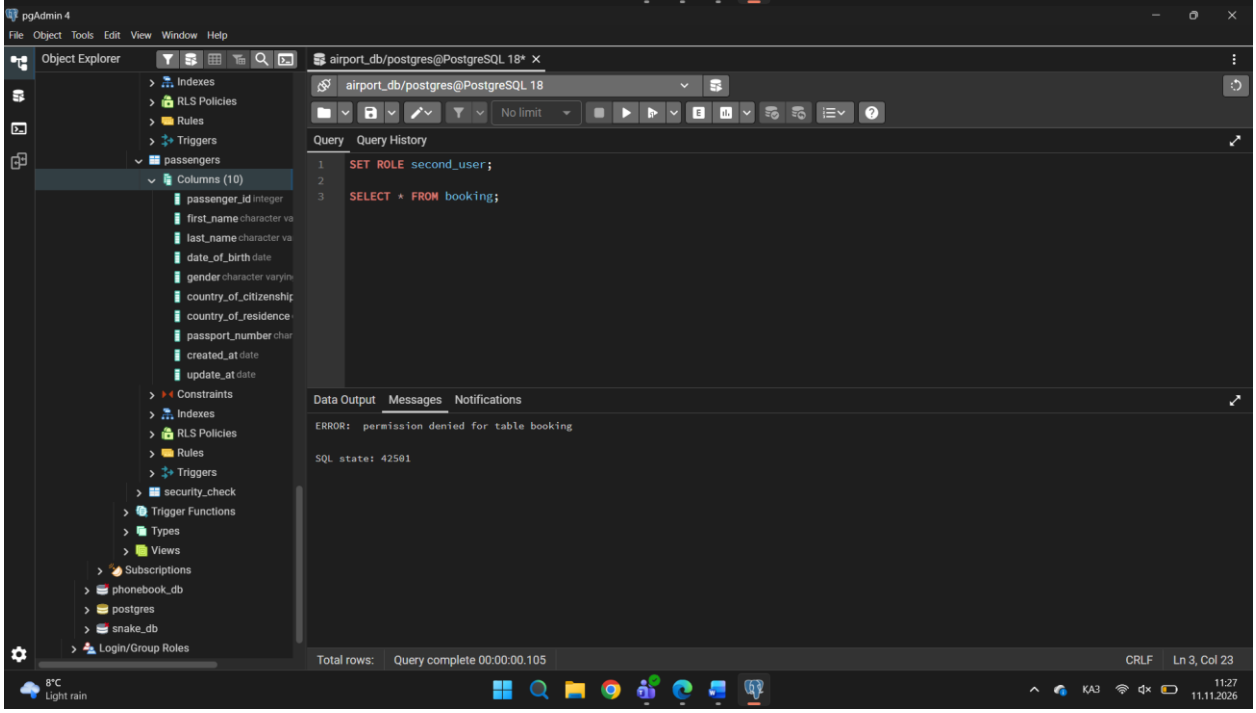
17. Grant column-level SELECT on only the full name and country columns от passengers.

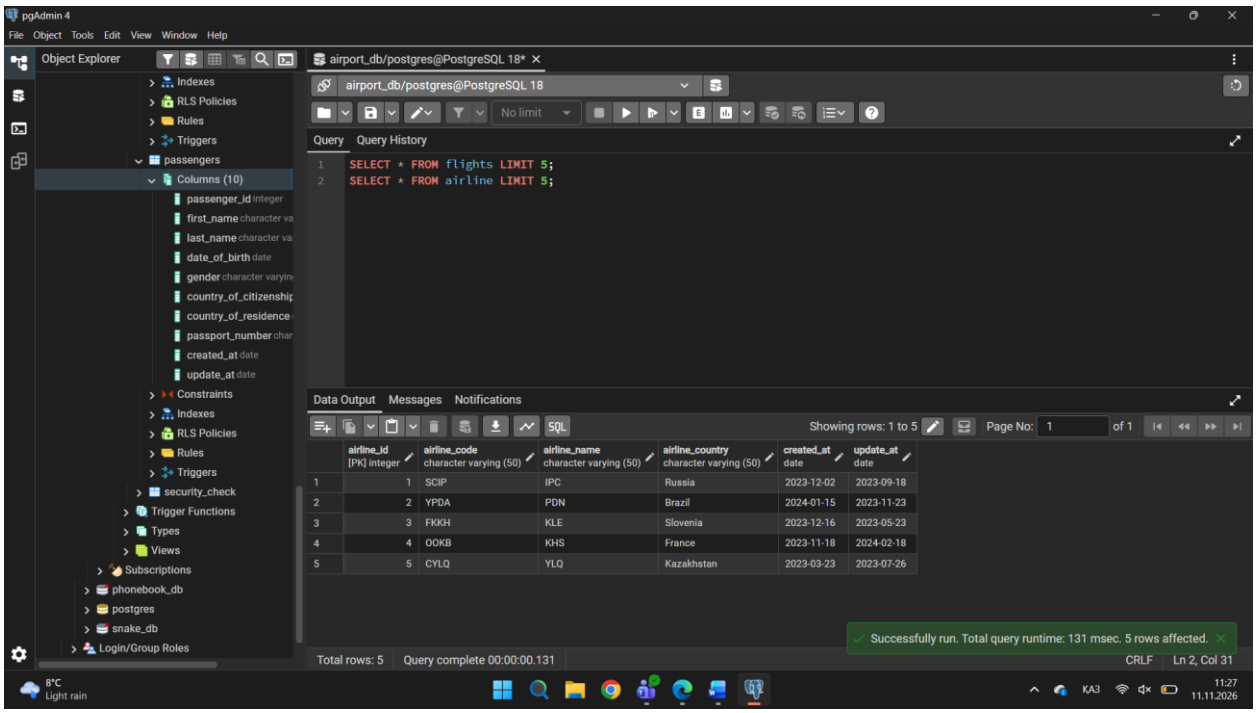




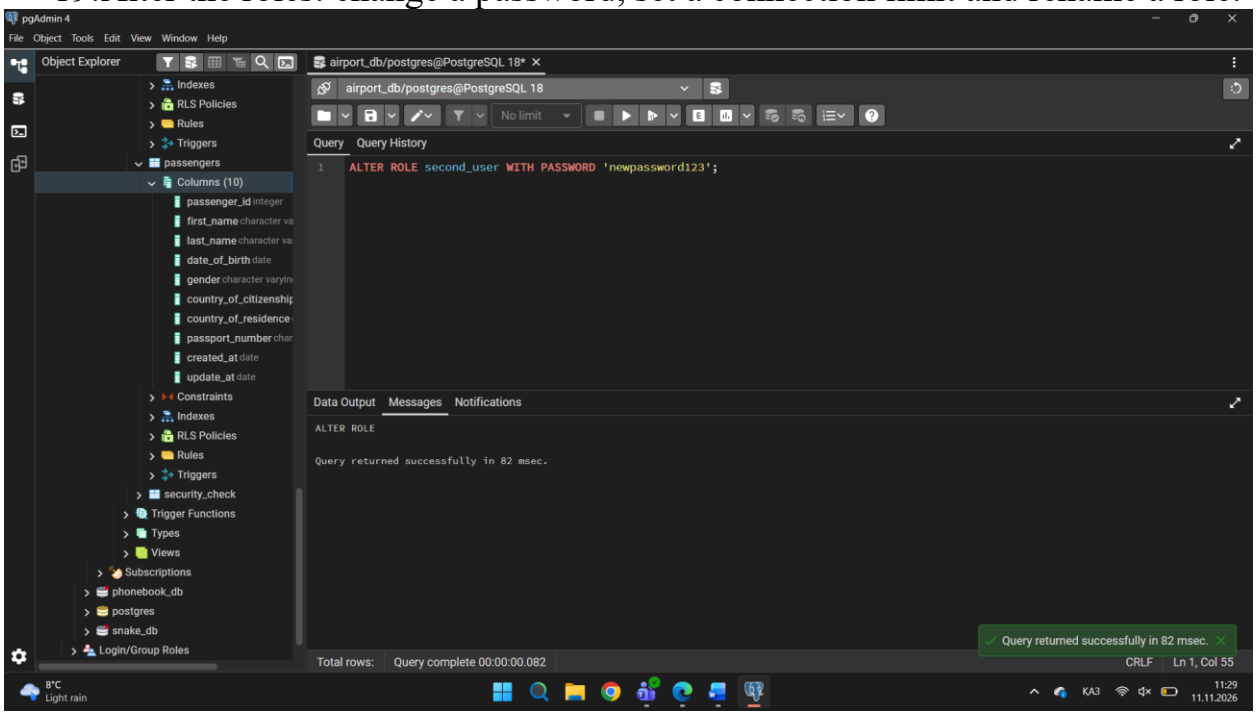
18.Revoke SELECT on bookings from flight_analyst and show that access is denied.

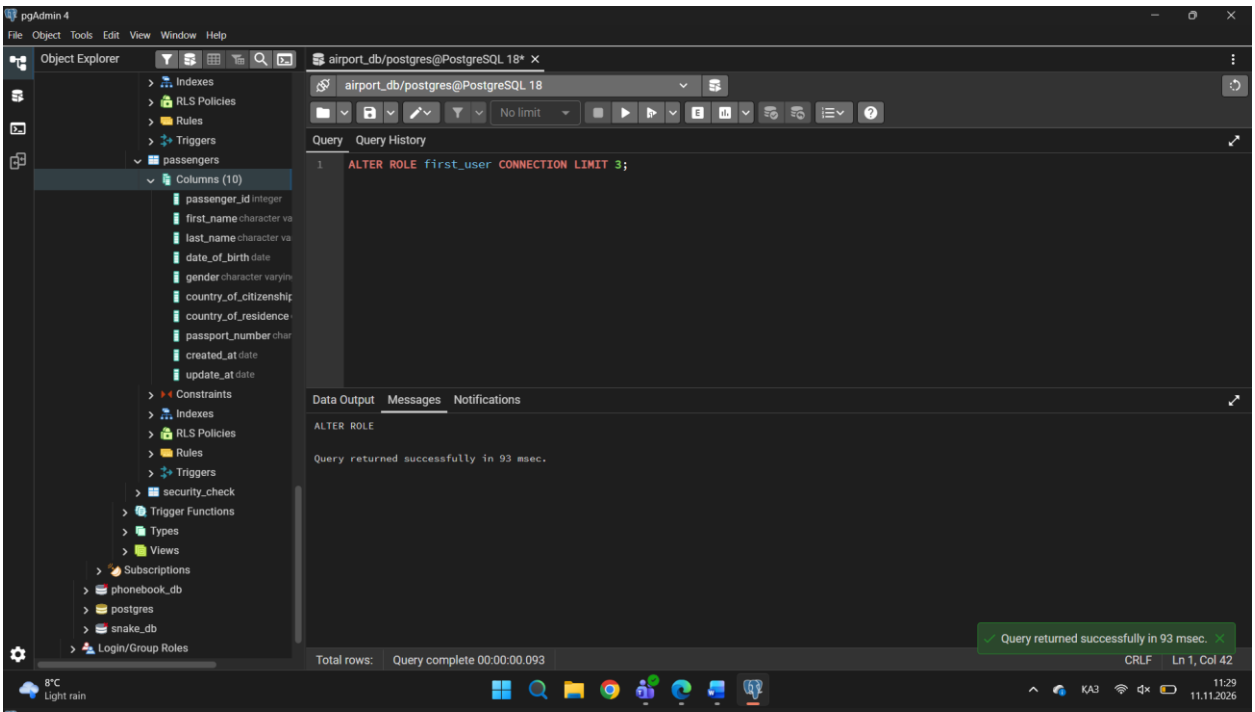

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 18*
airport_db/postgres@PostgreSQL 18
No limit
Query Query History
1 REVOKE SELECT ON booking FROM flight_analyst;
Data Output Messages Notifications
REVOKE
Query returned successfully in 93 msec.
Query returned successfully in 93 msec.
Total rows: Query complete 00:00:00.093
CRLF Ln 1, Col 46
8°C Light rain
11:27 11.11.2026

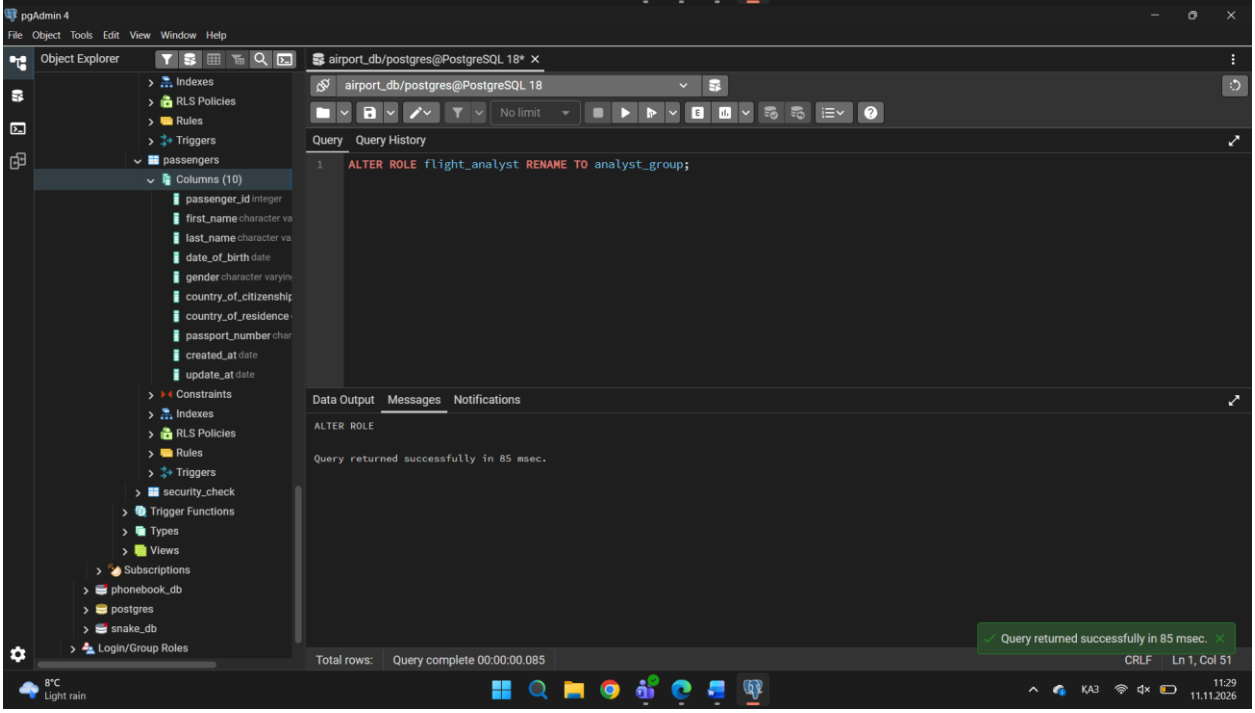

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 18*
airport_db/postgres@PostgreSQL 18
No limit
Query Query History
1 SET ROLE second_user;
2
3 SELECT * FROM booking;
Data Output Messages Notifications
ERROR: permission denied for table booking
SQL state: 42501
Total rows: Query complete 00:00:00.105
CRLF Ln 3, Col 23
8°C Light rain
11:27 11.11.2026

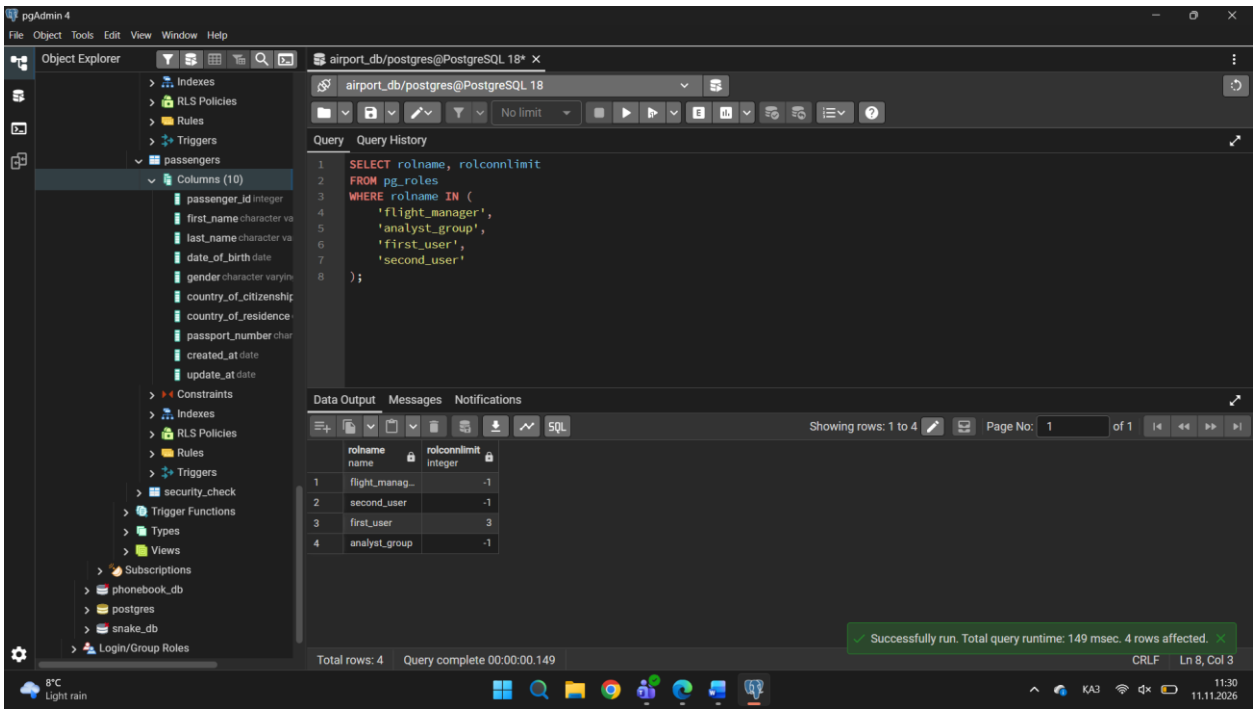


19.Alter the roles: change a password, set a connection limit and rename a role.

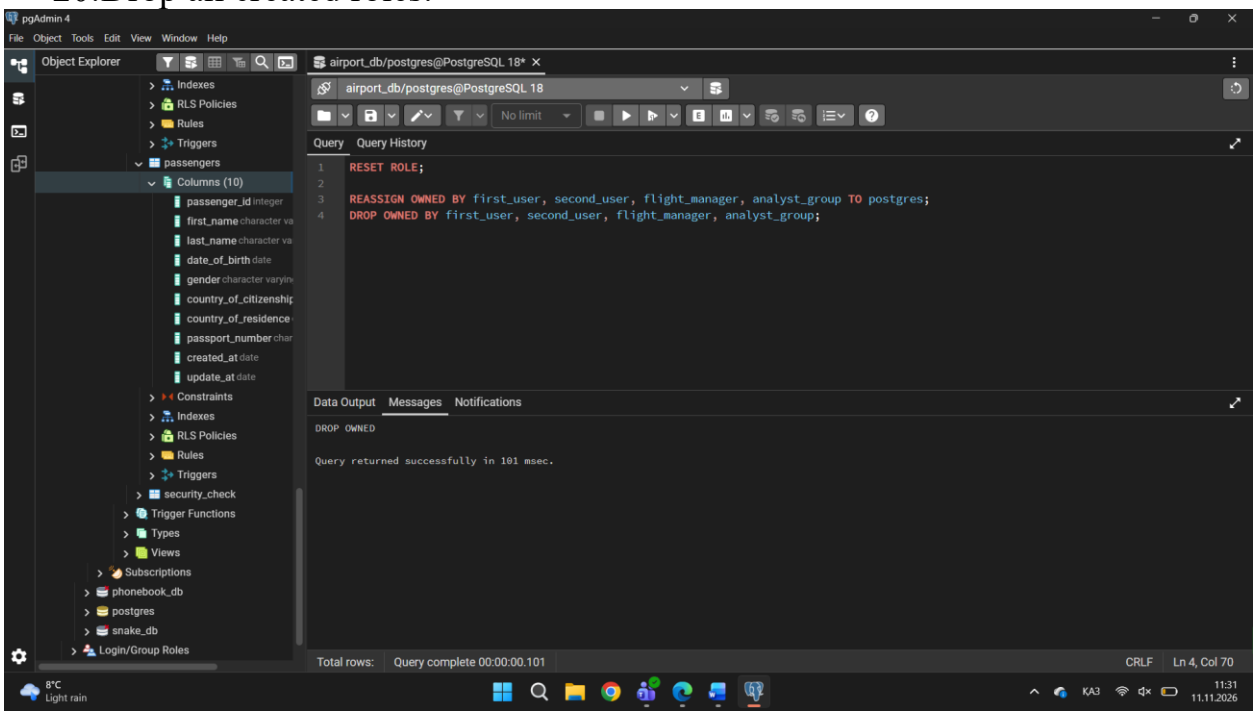


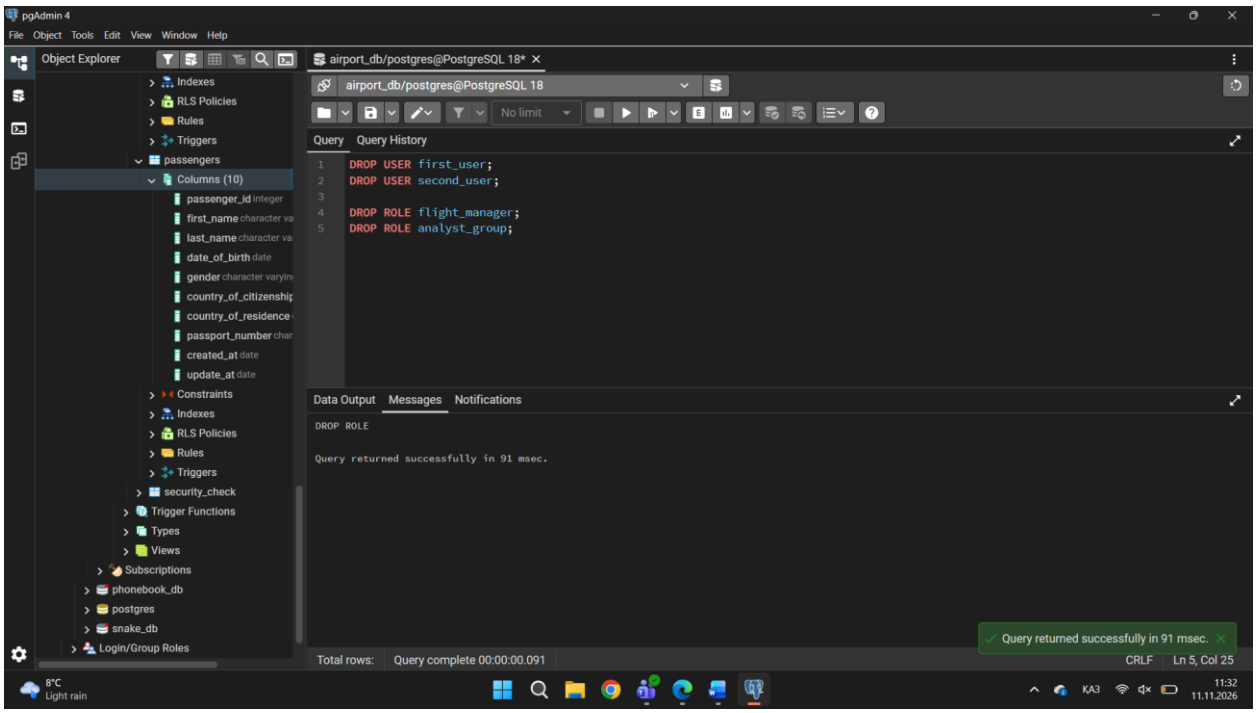

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 18*
airport_db/postgres@PostgreSQL 18
No limit
Query Query History
1 ALTER ROLE first_user CONNECTION LIMIT 3;
Data Output Messages Notifications
ALTER ROLE
Query returned successfully in 93 msec.
Query returned successfully in 93 msec.
Total rows: Query complete 00:00:00.093
CRLF Ln 1, Col 42
8°C Light rain
11:29 11.11.2026


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 18*
airport_db/postgres@PostgreSQL 18
No limit
Query Query History
1 ALTER ROLE flight_analyst RENAME TO analyst_group;
Data Output Messages Notifications
ALTER ROLE
Query returned successfully in 85 msec.
Query returned successfully in 85 msec.
Total rows: Query complete 00:00:00.085
CRLF Ln 1, Col 51
8°C Light rain
11:29 11.11.2026

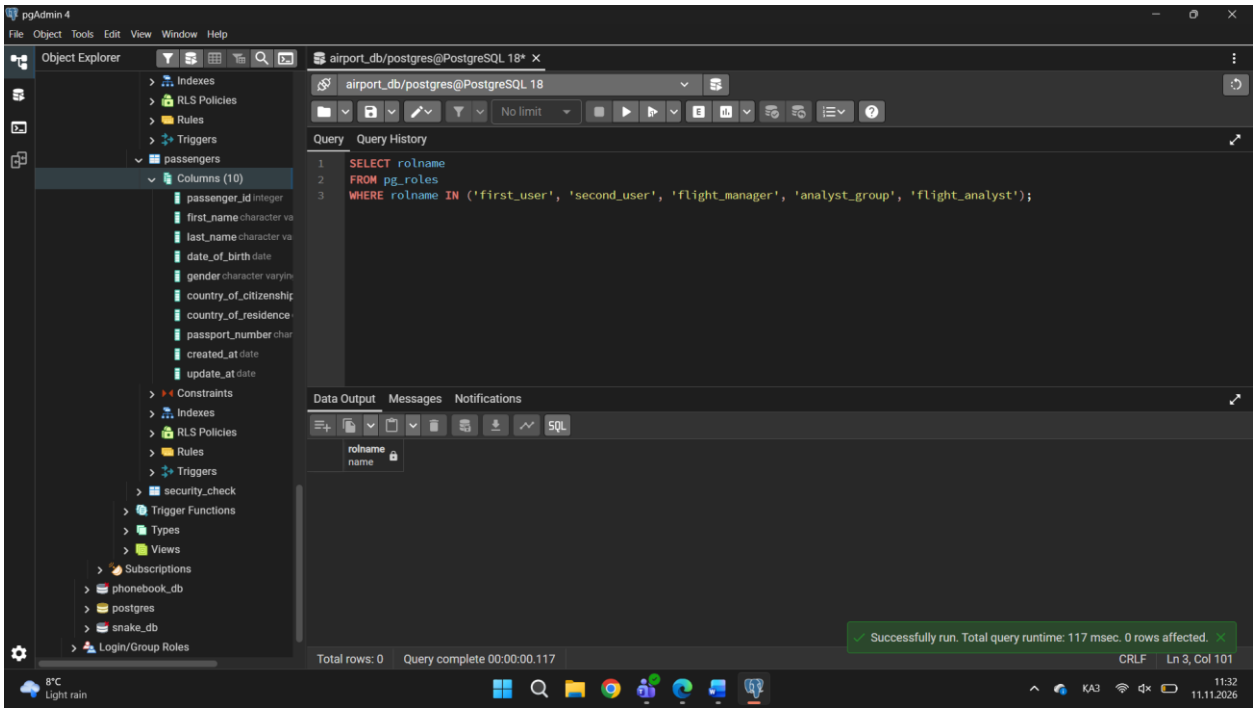


20.Drop all created roles.





Checking



Omar Aknaz, 11.11.2026 DataBases